

Reasoning Method Spectrum: From Low Cost to High Reasoning Quality

□□□□□ [Wei et al., 2022] [Wang et al., 2023] [Yao et al., 2023]

Standard Prompting

CoT

Self-Consistency

ToT / Search

Compute Cost ↗

Low

High

Standard Prompting

Single generation, no reasoning

GSM8K: 17.9%

□□: 1x

For: Simple classification

CoT

Single-Path Reasoning

GSM8K: 56.9%

□□: 1.5x

For: Math reasoning

Self-Consistency

Multi-Path Voting

GSM8K: 74.4%

□□: 10-40x

For: Complex reasoning

ToT / □□

BFS/DFS exploration

24-Game: 74%

□□: $O(k^d)$

For: Planning & search